Financial and Ratio Analysis

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Financial Analysis

- The purpose of financial analysis is to assess the performance of a firm <u>in the context of its stated goals and strategy.</u>

- **Two** primary tools:
  - **Ratio analysis:** to assess how various line items in financial statements relate to each other and to measure relative performance.
  - **Cash flow analysis** to evaluate liquidity and the management of operating, investing, and financing activities as they relate to cash flow.

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Determinants of firm value and ratio analysis

- Profitability and growth drive firm value.
- Managers can employ four levers to achieve growth and profit targets:
  - Operating management
  - Investment management
  - Financing strategy
  - Dividend policy
- Ratio analysis seeks to evaluate the firm's effectiveness in these areas.

Ratio Analysis

- Financial statements provide absolute amounts, but "good or bad" performance is relative
- Evaluating ratios requires comparison against some benchmark.
  - Ratios over time from prior periods (time-series)
  - Ratios of other firms in the industry (cross-sectional)
  - Some absolute benchmark (e.g., cost of capital)
- Note: All ratios need to be calculated after standardising financial statements and accounting adjustments (See "HM06-20 spreadsheet" on Moodle)

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Ratio Analysis (cont'd)

- No standards for ratio definitions
- Ratio analysis helpful for analysing
  - Performance/Profitability/Growth
  - Efficiency of utilising resources
  - Liquidity
  - Long-term solvency
- Limitations
  - They can be manipulated by timing the firm's activities appropriately or changing accounting estimates
  - That's why we make our own accounting adjustments and recast financial statements

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Ratio Analysis 1: Profitability

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Measuring Overall Profitability

- Return on Equity (ROE): measures how much (accounting) return a firm generates for its shareholders

$$\text{ROE} = \frac{\text{Profit or loss}}{\text{Shareholders' Equity}}$$

- Benchmarks for ROE
  - Previous years' ROE
  - Industry peers' ROE
  - Cost of equity capital
- Starting point for financial analysis

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Decomposing ROE: DuPont Approach

$$\text{ROE} = \frac{\text{Profit or loss}}{\text{Total assets}} \times \frac{\text{Total assets}}{\text{Shareholders' equity}}$$

$$= \text{Return on Assets (ROA)} \times \text{Equity multiplier}$$

$$\text{ROA} = \text{Net Profit Margin} \times \text{Asset Turnover}$$

$$= \frac{\text{Profit or loss}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Total assets}}$$

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Decomposing ROE: DuPont Approach (cont'd)

ROE

$$= \frac{\text{P\&L}}{\text{S/H Equity}}$$

Profit Margin

$$= \frac{\text{P\&L}}{\text{Sales}}$$

Net Profit Margin (NPM)

X

Efficiency

$$= \frac{\text{Sales}}{\text{Total Assets}}$$

Asset Turnover (ATO)

X

Leverage

$$= \frac{\text{Total Assets}}{\text{Total Equity}}$$

Asset to Equity (ATE)

Mini Case: International apparel retailer

H&M

- Sweden-based
- Publicly listed in Stockholm Stock Exchange
- 101,000 employees
- 4,300+ stores in 79 countries
- €20b revenue in 2023

Inditex

- Spain-based
- Publicly listed in Spanish Stock Exchange
- 160,000 employees
- 5,700 stores in 200+ countries
- €36b revenue in 2023

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Case: H&M vs. Inditex ROE

Higher profit margin drives higher ROA of Inditex relative to H&M, even though H&M had higher asset turnover in 2020.

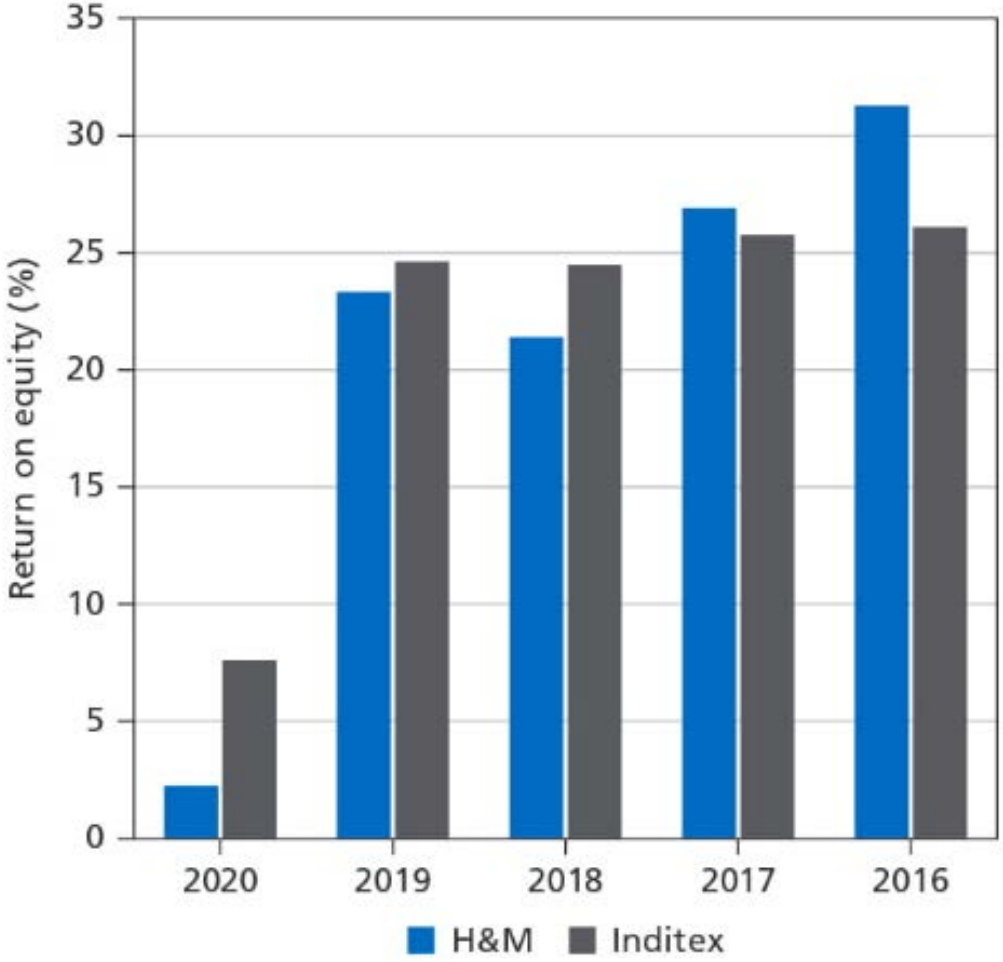
Despite higher leverage, H&M still has lower ROE due to very low ROA in 2020.

	H&M		Inditex	
Ratio	**2020**	**2019**	**2020**	**2019**
Net profit margin (ROR) (%)	0.7	5.8	5.4	12.9
× Asset turnover	1.04	1.26	0.74	1.03
= Return on assets (ROA) (%)	0.7	7.3	4.0	13.3
× Equity multiplier	3.23	3.21	1.86	1.85
= Return on equity (ROE) (%)	2.2	23.3	7.5	24.6

Who has better profitability in 2019 and 2020?

What are the key drivers of their profitability?

ROE for H&M and Inditex



Limitations of ROA

- P&L includes
  - Profit from operating <u>and</u> investing activities
  - Interest income and expenses (financing activities)
- To measure <u>only</u> *operating performance*, we use **<u>Return on Net Operating Assets (RNOA)</u>**

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Return on Net Operating Assets (RNOA)

$$\text{RNOA} = \frac{\text{Net Operating Profit After Tax (NOPAT)}}{\text{Net Operating Assets}}$$

- Measures profit generated from operating activities given the amount of operating assets employed
- Excludes impact from financing and investing activities

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Net Operating Profit After Tax (NOPAT)

NOPAT = P&L – Net investment profit after tax + Interest expense after tax

= (Investment income + Interest income) × (1 – Tax rate)

= Interest expense × (1 – Tax rate)

- Excludes: income from investment and interest – investing and financing activities
- Add back: interest expense – financing activities
- All obtained from income statement

An alternative way:

NOPAT = (Revenue-Operating expenses) × (1 – Tax rate)

Net Operating Assets

Treated as investment assets: defined as total cash minus the cash balance needed for operations (we assume the latter is 5% of revenue in H&M's case, i.e., the long-term average cash balance in European apparel retail industry).

Net Operating Assets = Net working capital + Net non-current operating assets

=

(Current assets – Excess cash and cash equivalents) – (Current liabilities – Current debt and current portion of non-current debt)

=

Non-current operating assets – Non-interest bearing non-current liabilities

- Exclude: excess cash and financial assets – investing activities
- Only keep cash needed for operations as operating assets
- Non-current operating assets generally include PP&E, intangible assets, and derivatives (for hedging)
- Non-interest bearing non-current liabilities generally include net deferred tax liability
- Exclude: interest-bearing debt (e.g., pension liabilities, finance lease liabilities, bank loans, bonds) – financing activities
- All obtained from balance sheet

Decomposing RNOA

$$\text{RNOA} = \frac{\text{Net Operating Profit After Tax (NOPAT)}}{\text{Net Operating Assets}}$$

$$= \frac{\text{NOPAT}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Net Operating assets}}$$

$$= \text{Net Operating Profit Margin} \times \text{Operating asset turnover}$$

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Example: H&M vs. Inditex RNOA

Who has better profitability from operating activities in 2019 and 2020? What are the key drivers of their RNOA?

	H&M		Inditex	
Ratio	**2020**	**2019**	**2020**	**2019**
Net operating profit margin (%)	1.1	6.2	5.7	13.1
× Net operating asset turnover	1.41	1.66	1.44	2.03
= Return on net operating assets (%)	1.6	10.3	8.2	26.5

- H&M has much lower NOPAT margin: stronger focus on lower-priced segment of the apparel industry.
- H&M also has lower operating asset turnover, although the gap is lower in 2020 (due to Covid19).

Discussion of Results from Profitability Analysis

For 2020	**H&M** Total Assets	**Inditex** Total Assets	**H&M** Operating Assets	**Inditex** Operating Assets
Asset Turnover	1.04	0.74	1.41	1.44
Return	0.7%	4.0%	1.6%	8.2%

<u>What do we learn from the above comparison?</u>

- H&M has higher efficiency over total assets than Inditex, but Inditex has higher efficiency over operating assets.
- Inditex has higher returns on both operating assets and total assets.

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Income Statement Ratios

- Common-sized income statements facilitate comparisons of key line items across time and different firms.
  - Express all line items as % of revenue
- Additionally, the following ratios are also helpful:
  - Gross profit margin
  - EBITDA margin

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$$\text{Gross profit margin} = \frac{\text{Revenue - Cost of Sales}}{\text{Revenue}}$$

- Gross profit margin measures the profitability of sales, less direct costs of sales and is an indicator of:
  - The price premium that a firm's product commands in the market
  - The efficiency of a firm's procurement and/or production process

$$\text{EBITDA margin} = \frac{\text{Earnings before interest, taxes, depn \& amort}}{\text{Revenue}}$$

- EBITDA margin eliminates the significant non-cash expenses of depreciation and amortization along with interest and taxes

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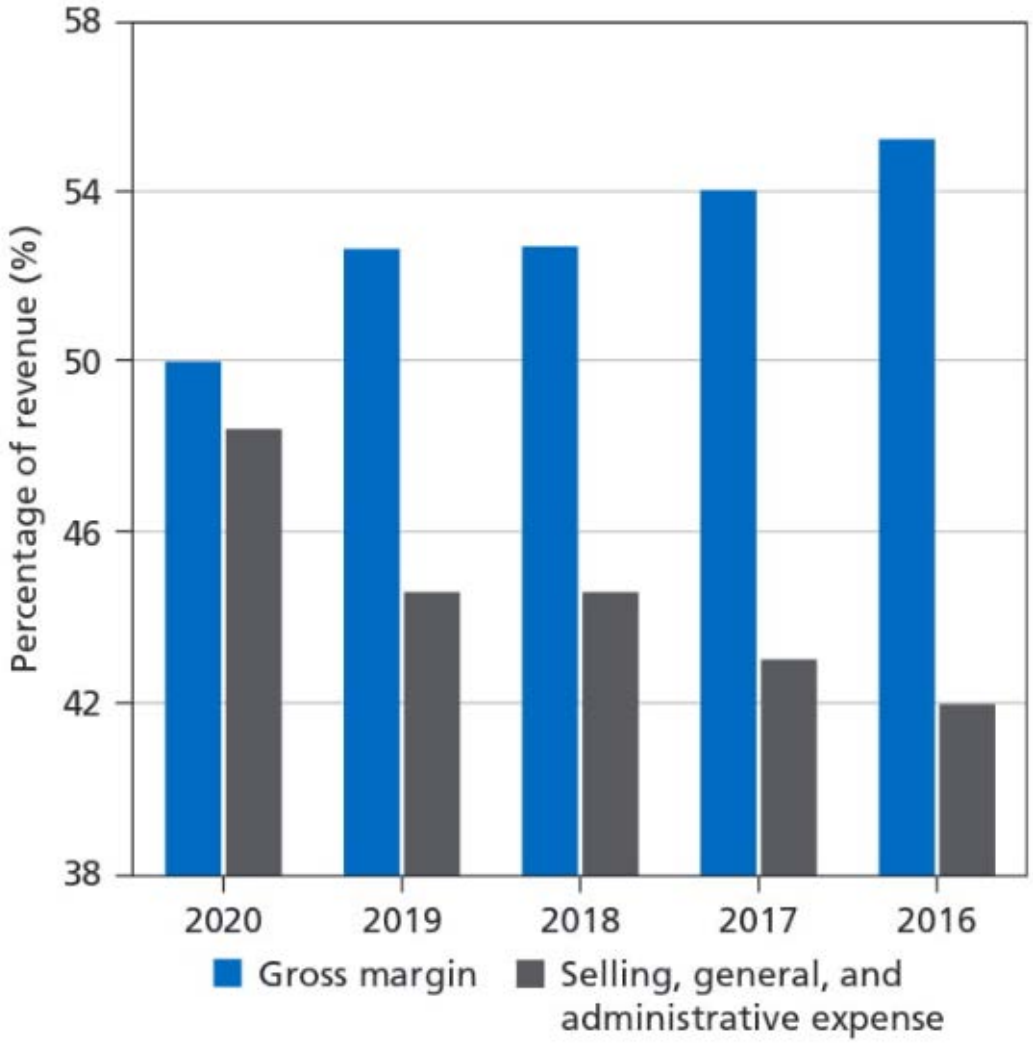
H&M income statement ratios



Ratio Analysis 2:
Efficiency of utilizing resources

Decomposing Asset Turnover

- Asset management is a key indicator of how effective a firm's management is in utilizing its resources
  - The focus is not on profits but on the <u>level of resources needed to maintain a given level of operations</u> (normally, a given level of sales)
- Asset turnover may be broken into two primary components:
  - Working capital management
  - Non-current asset management

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Working Capital Management

- Efficiency of utilizing a firm's working capital

Net working capital = (Current assets – Excess cash and cash equiv.)
– (Current liabilities – Current debt and current portion of non-current debt)

- Key ratios include:
  - Net working capital to sales
  - Net working capital turnover
  - Accounts receivable turnover or Days' receivables
  - Inventory turnover or Days' inventory
  - Accounts payable turnover or Days' payables
  - Cash conversion cycle

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$$\text{Net working capital to sales ratio} = \frac{\text{Net working capital}}{\text{Revenue}}$$

$$\text{Net working capital turnover} = \frac{\text{Revenue}}{\text{Net working capital}}$$

$$\text{Accounts receivables turnover} = \frac{\text{Revenue}}{\text{Accounts receivables}}$$

More accurately capturing inventory cost.

$$\text{Inventory turnover} = \frac{\text{Cost of sales}}{\text{Inventories}} \text{ or } \frac{\text{Cost of materials}}{\text{Inventories}}$$

$$\text{Accounts payables turnover} = \frac{\text{Cost of sales}}{\text{Accounts payables}} \text{ or } \frac{\text{Cost of materials}}{\text{Accounts payables}}$$

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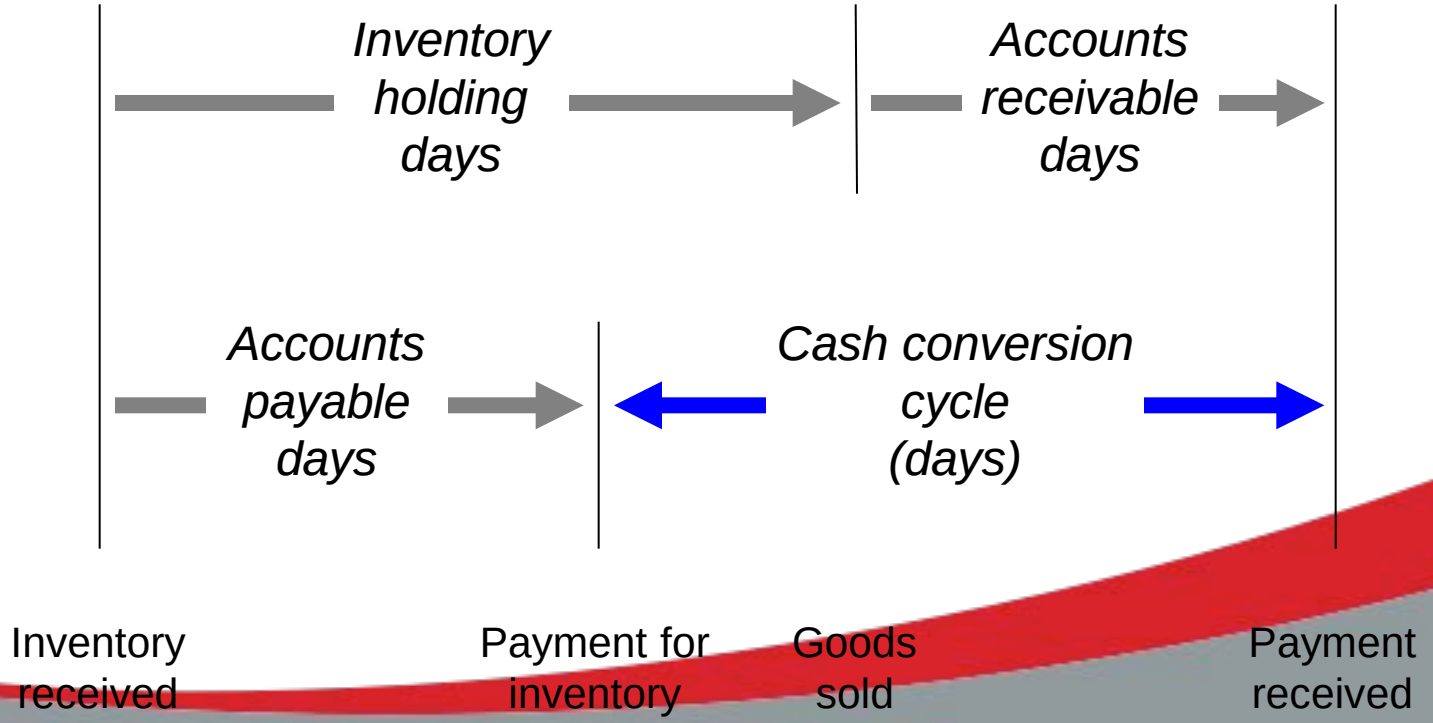
$$\text{Days' receivables} = \frac{\text{Days in period}}{\text{Accounts receivables turnover}}$$

$$\text{Days' inventory} = \frac{\text{Days in period}}{\text{Inventory turnover}} \qquad \text{Days' payables} = \frac{\text{Days in period}}{\text{Accounts payable turnover}}$$

$$\text{Cash conversion cycle} = \text{Days' inventory} + \text{Days' receivables} - \text{Days' payables}$$



Non-Current Assets Management

- Efficiency of utilizing a firm's long-term assets
- Key ratios include:

$$\text{Net non current operating asset turnover} = \frac{\text{Revenue}}{\text{Net non current operating assets}}$$

$$\text{PP\&E Turnover} = \frac{\text{Revenue}}{\text{Net property, plant, and equipment}}$$

  – Measure the amount of sales generated by £1 invested

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H&M was less efficient at managing working capital in 2020, driven by lower inventory turnover and receivables turnover: a result of Covid19

Who is more efficient at managing assets?

Inditex has very high inventory turnover – fast fashion, own manufacturing, local suppliers, centralized logistics centre, although slower in 2020 due to Covid19.

	H&M		Inditex	
Ratio	**2020**	**2019**	**2020**	**2019**
Operating working capital/Revenue (%)	11.6	9.9	–3.0	–1.9
Net non-current operating assets/Revenue (%)	59.3	50.4	72.6	51.1
PP&E/Revenue (%)	53.1	46.3	66.8	49.2
Operating working capital turnover	8.59	10.10	–33.57	–53.55
Net non-current operating asset turnover	1.69	1.98	1.38	1.96
PP&E turnover	1.88	2.16	1.50	2.03
Trade receivables turnover	41.72	38.13	84.83	124.33
Days' receivables	8.6	9.4	4.2	2.9
Inventories turnover	2.41	2.91	3.93	4.63
Days' inventories	149.5	123.9	91.7	77.8
Trade payables turnover	10.55	14.99	2.43	3.15
Days' payables	34.1	24.0	148.2	114.1
Cash conversion cycle (in days)	124.0	109.3	–52.3	–33.4

Inditex pays its suppliers much later, i.e., financing its working capital by supplier credit. Strong bargaining power with suppliers.

Financial Leverage Analysis

- Borrowing allows a firm to access capital but increases the risk of ownership for equity holders.
- Analysis of leverage can be performed on both current and non-current debt:
  - Liquidity analysis relates to evaluating current liabilities
  - Solvency analysis relates to longer term liabilities
- Measure a firm's ability to meet its obligations to creditors

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Ratio Analysis 3:
Short-term liquidity

Liquidity Analysis

- Measure a firm's ability to pay its current obligations.
- Key ratios include:
  - Current ratio, Quick ratio, and Cash ratio (a firm's ability to pay current liabilities out of near-cash assets)
  - Operating cash flow ratio (ability of a firm's operations to pay current liabilities)

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Liquidity Analysis (cont'd)

$$\text{Current ratio} = \frac{\text{Current assets}}{\text{Current liabilities}}$$

$$\text{Quick ratio} = \frac{\text{Cash and marketable securities} + \text{ Accounts receivable (net)}}{\text{Current liabilities}}$$

$$\text{Cash ratio} = \frac{\text{Cash and marketable securities}}{\text{Current liabilities}}$$

$$\text{Operating cash flow ratio} = \frac{\text{Cash flow from operations}}{\text{Current liabilities}}$$

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H&M vs. Inditex - Liquidity Ratios

Ratio	H&M		Inditex	
	2020	2019	2020	2019
Current ratio	1.07	1.01	1.38	1.32
Quick ratio	0.32	0.30	0.93	0.80
Cash ratio	0.24	0.20	0.89	0.76
Operating cash flow ratio	0.45	0.82	0.44	1.08

- Both H&M and Inditex have comfortable liquidity situations in 2019 (current ratio>1).
- Both firms have significantly lower operating cash flow ratio in 2020 due to Covid19.
- H&M's current ratio, quick ratio and cash ratio increased slightly in 2020 – potentially due to paying no cash dividends.

Ratio Analysis 4:
Long-term solvency

Solvency analysis

- Beyond short-term survival, solvency measures the ability of a firm to meet long-term obligations.

$$\text{Liabilities-to-equity ratio} = \frac{\text{Total liabilities}}{\text{Shareholders' equity}}$$

$$\text{Debt-to-equity ratio} = \frac{\text{Current debt} + \text{Non-current debt}}{\text{Shareholders' equity}}$$

$$\text{Debt-to-capital ratio} = \frac{\text{Current debt} + \text{Non-current debt}}{\text{Current debt} + \text{Non-current debt} + \text{Shareholders' equity}}$$

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Solvency analysis (cont'd)

In US GAAP, CFO includes interest paid, interest received, and dividends received, while IFRS allows either including or excluding these iterms in CFO. If CFO does not include interest paid, then we do not add back interest paid.

- Two ratios that specifically address the ability to pay interest on debt are:

$$\text{Interest coverage (earnings-based)} = \frac{\text{Profit or loss} + \text{Interest expense} + \text{Tax expense}}{\text{Interest expense}}$$

$$\text{Interest coverage (cash flow-based)} = \frac{\text{Cash flow from operations} + \text{Interest paid} + \text{Taxes paid}}{\text{Interest paid}}$$

- Measure the ease with which firm meets interest payments

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Comparison of H&M and Inditex Debt and Coverage Ratios

	H&M		Inditex	
Ratio	2020	2019	2020	2019
Liabilities-to-equity	2.23	2.21	0.86	0.85
Debt-to-equity	1.48	1.47	0.46	0.41
Debt-to-invested capital	0.60	0.59	0.31	0.29
Interest coverage (earnings-based)	2.57	11.51	11.95	31.01
Interest coverage (cash flow-based)	23.27	32.28	44.98	67.39

- H&M has much higher leverage than Inditex: a more aggressive debt policy.
- H&M has lower interest coverage than Inditex, due to higher debt
- Both firms have sufficient earnings and cash flow to cover interest expenses.

Assessing the Sustainable Growth Rate

- A comprehensive measure of a firm's ratios is the sustainable growth rate:

$$\text{Sustainable growth rate} = \text{ROE x (1 - Dividend payout ratio)}$$

$$\text{Dividend payout ratio} = \frac{\text{Cash dividends paid}}{\text{Profit or loss}}$$

- Sustainable growth rate is the rate at which a firm can grow while maintaining its profitability and financial policies.

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Sustainable Growth Rates For H&M and its peers

	H&M		Inditex	
Ratio (%)	2020	2019	2020	2019
ROE	2.2	23.3	7.5	24.6
Dividend payout ratio	0.0	120.0	98.7	75.2
Sustainable growth rate	2.2	−4.7	0.1	6.1

Whose growth is more sustainable? Why?

- In 2020, Inditex has higher ROE but pays much higher dividends and thus has lower sustainable growth rate than H&M.
- In 2019, H&M paid very high dividends, thus negative sustainable growth rate.

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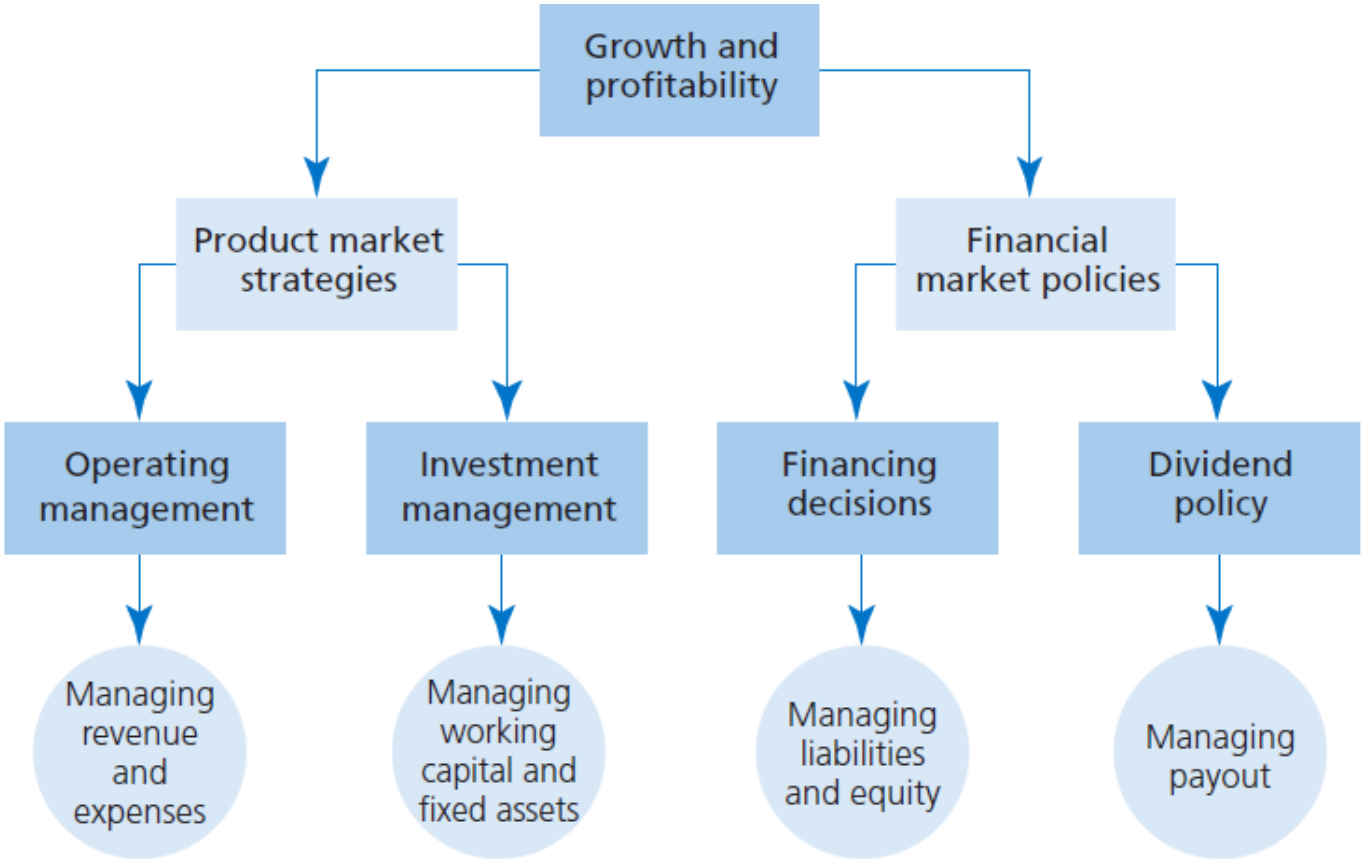
Drivers of Firms' Profitability and Growth



Notes to Ratio Analysis

- When using <u>*balance sheet*</u> numbers to calculate ratios, for better accuracy, we use ***the average value***, i.e., $\frac{Beginning\ balance + Ending\ Balance}{2}$

  E.g., ROA=Profit or loss/Average total assets

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Condensed Financial Statements

- See H&M spreadsheet as an example.
  - Worksheet “Adjusted statements”, Row 95-149
- You are NOT required to know how to prepare condensed financial statements.
- But, you need to know how to read them.
  - Some ratios are calculated using numbers from condensed financial statements
  - E.g., ROE=Profit or loss (to group)/Group equity

Condensed Financial Statements (cont'd)

- Profit or loss (to group)– Income Statement
  - Measures profits to the whole group
  - Conceptually, it includes Profit/Loss to Ordinary Shareholders, and Profit/Loss to Non-Controlling Interest.

- Group equity – Balance Sheet
  - Measures shareholders' equity of the entire group
  - Conceptually, it includes Ordinary Shareholders' Equity and Minority Interest

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Condensed Financial Statements (cont'd)

- <u>Net Assets Held for Sale</u>

  = Assets Held for Sale – Liabilities Held for Sale

  IFRS requires assets/liabilities held for sale to be reported separately, neither current nor non-current

  Group equity = Ordinary shareholders' equity + Minority interest – Net assets held for sale

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Condensed Financial Statements (cont'd)

- <u>Income statement:</u>

  Net business profit after tax = Net operating profit after tax + Net investment profit after tax

  Profit or loss = Net business profit after tax – Net interest expense after tax

- <u>Balance sheet:</u>

  Business assets = Net operating assets+ Net investment assets†

  Capital=Debt + Group equity

† Net investment assets=Excess cash + Minority equity investment + Other non-operating investments

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Exercise

- Download Excel file “H&M 06-20 spreadsheet-student template” on Moodle.
- Use worksheet “Adjusted Statements” to calculate the all the ratios between 2007 and 2020.
- Use the template given under worksheet “Ratio Analysis”)

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Conclusions

- The primary tool in financial analysis is **Ratio analysis**
  - To assess how various line items in financial statements relate to each other and to measure relative performance.
  - It must be evaluated while considering whether firm performance is consistent with the strategic initiatives of management.

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